# DMML-Net: Deep Metametric Learning for Few-Shot Geographic Object Segmentation in Remote Sensing Imagery

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***Abstract*— Geographic object segmentation is a fundamental yet challenging problem for remote sensing image interpretation. The prevalent paradigm to solve this problem is to train deep neural networks on massive labeled samples. Although remarkable achievements have been attained, these methods suffer from the severe dependence on the large-scale dataset and require a long training process with high computation burden. To address these issues, a deep metametric learning framework, named DMML-Net, consisting of the metametric learner and the base-metric learner, is proposed for few-shot geographic object segmentation. First, DMML-Net formulates the segmentation as the metric-based pixel classification and develops a deep feature pyramid comparison network as the architecture of the metric learner for multiscale metric learning. Benefiting from this design, the segmentation can be efficiently solved, as well as being robust to deal with the scale variations of geographic objects. Second, an affinity-based fusion mechanism is introduced to adaptively reweight and fuse the semantic information across samples, effectively calibrating the deviation of prototypes induced by the intraclass variations. Third, considering the impact of the large interclass distribution divergences, DMML-Net presents a metametric training paradigm to provide the metric model with flexible scalability for fast adaptation to novel tasks. After metatraining, DMML-Net can be applied for the few-shot segmentation tasks of novel geographic objects with only a few gradient steps on the small training set. Experimental results on two benchmark remote sensing datasets demonstrate the validity and the superiority of our method in low-shot conditions where there are only one to ten labeled samples.**

***Index Terms*— Few-shot learning, few-shot segmentation, metalearning, metric learning, semantic segmentation.**

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## I. Introduction

GEOGRAPHIC object segmentation, which focuses on assigning each pixel in the image with one of the predefined geospatial object categories, is an important component for the automatic processing and analysis of remote sensing data. It is widely used in a variety of applications, such as land use, urban management, environment monitoring, and other national economic construction fields [1]–[3].

Over the past decades, extensive efforts have been made for remote sensing geographic object segmentation, and a detailed review can be found in [4]–[6]. Especially, in recent years, from the milestone work of fully convolutional networks (FCNs) [7] for semantic segmentation on natural images, a variety of convolutional neural network (CNN)-based methods [8]–[11] have been proposed and dominated the field of remote sensing geographic object segmentation.

Despite the success achieved by the CNN-based methods, one common issue lies in the heavy dependence on plentiful annotated data since the deep CNN architectures usually contain massive parameters to be optimized. However, considering the prohibitive cost for data annotation and the scarcity of target sources, the total number of available training data in the real scenarios of the remote sensing field can be quite limited. In low data regimes, the existing CNN-based methods tend to fall into the problem of overfitting [12] and expose poor generalization ability. Meanwhile, a long training process with high computation is usually required for compatibility with the tasks of novel classes, which fails to satisfy the need for fast deployment. Therefore, it is necessary to explore effective learning algorithms to deal with the small data issues for geographic object segmentation in remote sensing imagery. Few-shot learning [13]–[15], a wide family of techniques that focuses on enhancing the efficiency of learning through experience, is one of the most promising approaches to address the small data issues. Specific to the field of few-shot semantic segmentation, most of the current relevant research work [16], [17] inherit the metric-based learning framework [18] due to its simplicity and efficiency. Concretely, the main idea is to utilize CNN to parameterize a metric space that is transferable across tasks. The metric space is expected to provide well separability for heterogeneous pixels, as well as keeping the compactness of the homogeneous pixels. Hence, the segmentation task can be resolved via the distance computation between the pixel

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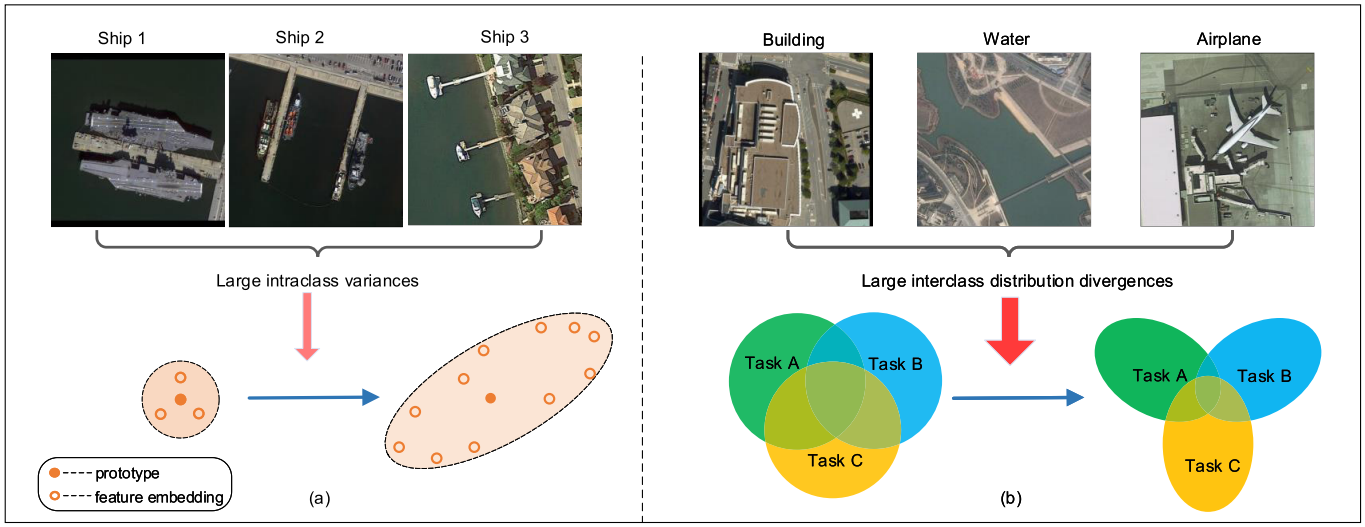



Fig. 1. Illustration of the two main factors that obstacle the metric learning for few-shot geographic object segmentation: (a) large intraclass variances and (b) large interclass distribution divergences.

embeddings and the class-specific prototype representations. This metric-based learning paradigm demonstrates the superiority over the few-shot segmentation of common objects in natural scenes. However, the remote sensing images are quite different from natural images because of their unique properties, and directly taking the few-shot learning methods of natural images for few-shot remote sensing image interpretation may lead to severe inappropriateness. The specific explanations are given as follows.

### A. Large Intraclass Variances

Considering the volatility of the imaging conditions, geographic objects of the same class may demonstrate visual differences in terms of multiple aspects, such as scales, colors, and shapes. As shown in Fig. 1(a), the increase in intraclass variances leads to the dilation and dispersion of the intraclass feature space, which becomes one obstacle for the estimation of the prototypes. Especially in low data regimes where the training set only contains very limited instances for the target category, the prototypes can be easily interrupted by the irrelevant noise features. Moreover, the metric model trained from the small dataset may generalize poorly when dealing with the novel instances that expose large appearance differences with the closed training set.

### B. Large Interclass Distribution Discrepancy

Concretely, there are obvious divergences in the visual characteristic of different geographic object classes, such as airplanes with regular shapes, water with vague boundaries, ships with small object scales, and buildings with large object scales. The optimal metric models for the segmentation tasks of various geographic object classes may have utterly different biases on the learning of feature descriptors. As illustrated in Fig. 1(b), with the increase in the divergences of interclass distribution, the sets of feature descriptors for different tasks become more discrete. Consequently, the metric knowledge, which is corresponding with the common feature descriptors across tasks, gets seriously weakened and exposes limited guidance value for the individual learning tasks.

To deal with the above issues, we propose a deep metametric learning framework, named DMML-Net, for the challenge of few-shot geographic object segmentation in remote sensing imagery. After metatraining, DMML-Net can be applied for the segmentation tasks of novel object classes with only a few gradient steps on the small training set. Three novel strategies are involved to construct DMML-Net. First, in light of the scale variability of geographic objects, a deep feature pyramid comparison network is developed as the architecture design of the metric learner to conduct multiscale metric learning. In particular, the prototypes of each class are initially extracted from the multiscale feature embeddings of the support set and then anisotropic rescaled for robust intraclass feature abstraction. The segmentation is efficiently solved in a contrastive way via the nonparametric nearest neighbor classification in the metric space. Second, an affinity-based fusion mechanism is designed to calibrate the deviation of the prototypes caused by the intraclass variances. Concretely, affinity is introduced to measure the semantic consistency across the samples of the same class. Based on the computed affinity scores, the semantic information of different support samples is adaptively reweighted and fused to enhance the representativeness of the estimated prototypes. Third, considering the impact of the large interclass distribution divergences, DMML-Net presents a novel metametric training paradigm. Specifically, DMML-Net adopts the episode-based strategy to construct the primary learning unit of the metric learner. The process of metric learning within each episode is explicitly decoupled into two-stage gradient optimization, including the task-specific adaptation and the cross-task knowledge accumulation. Compared with the conventional metric training methods, this metametric learning paradigm helps boost the metric learning efficiency and provides the model with flexible scalability for fast adaption to novel tasks. In addition, for effective training and evaluation of few-shot geographic object segmentation, a benchmark dataset, named mini-iSAID, is introduced based

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on the public dataset iSAID [19], which may provide convenience for future researches. Finally, the proposed approach is evaluated on two benchmark datasets, and the experimental results demonstrate that the proposed DMML-Net outperforms the existing baseline methods and achieves state-of-the-arts results in low data regimes where only one to ten labeled samples are available.

In summary, the main contributions of this work are given as follows.

1) We propose a deep metametric learning framework, named DMML-Net, for the challenge of few-shot geographic object segmentation in remote sensing images. DMML-Net can be applied for the novel few-shot segmentation tasks of unseen geographic object categories with only a few gradient steps, significantly reducing the data dependence and enabling fast learning ability for novel tasks.
2) DMML-Net tackles the segmentation as the metric-based pixel classification and presents a deep feature pyramid comparison network as the architecture of the metric learner, which is capable of extracting the scale-balanced prototypes from limited samples for multiscale dense metric comparison. This design helps increase the robustness of the model in the scale variations of geographic objects.
3) An affinity-based fusion mechanism is designed to adaptively reweight and fuse the semantic information from multiple samples, which effectively calibrates the deviation of the prototype estimation induced by the large intraclass variances.
4) A metametric learning paradigm is presented for the proposed DMML-Net. Both the task-specific and cross-task knowledge are jointly mined and leveraged to enhance the few-shot learning. Compared with the conventional metric learning methods, this approach demonstrates superior generalization ability over the novel tasks with large interclass distribution divergences.

The rest of this article is organized as follows. Section II gives a brief review of related works. Section III describes the proposed method in detail. Section IV presents the comprehensive analysis of the experimental results. Finally, Section V concludes this article and discusses future directions.

## II. Related Work

In this section, we review the work related to few-shot geographic object segmentation concerning three aspects: general semantic segmentation, few-shot learning, and semantic segmentation with limited labeled samples.

### A. General Semantic Segmentation

Semantic segmentation, which aims to assign a label from a set of predefined categories to every pixel in the image, is an important approach for refined image interpretation and has been widely investigated by various researchers. Recent advanced achievements mainly rely on deep CNNs. Long *et al.* [7] first propose the FCNs for semantic segmentation. FCN is widely favored by researchers for the merits of the powerful representation ability and end-to-end learning pattern, becoming the cornerstone of the following works. One prevalent research line focuses on the aggregation of multiscale contextual information. For example, PSPNet [20] proposes to embed the contextual information via the integration of different region-based feature representations. Similarly, DeepLab [21] introduces the atrous spatial pyramid pooling techniques to simultaneously capture the contextual information at multiple scales, which is efficiently implemented by applying atrous convolutions with different dilated rates in parallel. Another typical variant concentrates on enlarging the receptive field of filters to capture the long-range contextual information. Relevant work includes large kernel convolution [22], astrous convolution [23], and attention mechanism [24]–[26]. As for other types of variants, they mainly focus on enhancing the detailed information of the prediction results [27]–[29]. These methods generally attempt to combine the advantages of the extracted feature from multiple levels of the network to improve the segmentation quality.

Specific to the segmentation of remote sensing imagery, the deep learning-based methods also gain much popularity in the remote sensing community and obtain remarkable progress. An early research [30] was given by applying pretrained CNNs to remote sensing images for evaluating the generalization performance. In [31], an encoder–decoder architecture-based segmentation system is proposed to deal with the complex spatial and geometrical information in ultrahigh-resolution images. Extensive efforts have been devoted to the exploitation of spatially contextual information to increase the certainty of the generated prediction results. Maggiori *et al.* [43] propose to learn spatial features at multiple resolutions and boost the labeling accuracy by the aggregation of the detailed information from different levels. Mou *et al.* [33] further introduce the spatial and channel relation units to augment feature representations via the relation reasoning. Several studies further explore effective learning strategies to cope with the high-resolution properties of remote sensing images. Chen *et al.* [34] and Ding*et al.* [35] adopt the dual-stream network architecture to simultaneously leverage both the global and local feature descriptors to enhance the segmentation performance in the scenario of ultrahigh resolution. Meanwhile, considering the exhausted spadework on the input patch size selection, Nogueira *et al.* [36] exploit a multicontext learning paradigm to adaptively choose the best patch size for inference.

Compared with the above methods that need to be trained with plentiful labeled samples for full supervision, our model is capable of quickly adapting to novel categories with only a few annotated samples, which can effectively reduce the dependence on data quantity, as well as accelerating the training process to handle the variability of learning tasks.

### B. Few-Shot Learning

Few-shot learning, which aims at recognizing novel categories from very limited labeled samples, is currently one of the most popular topics in the machine learning community [37]–[39]. Vinyals *et al.* [40] introduce the episode-based

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learning schema to mimic the few-shot test scenarios during training for enhancing generalization. Snell *et al.* [18] and Sung *et al.* [41] investigate to extract prototype representations of each class by the nonlinear mapping in the metric space and conduct classification by the nearest neighbor classifier. Meanwhile, Ravi and Larochelle [42] analyze the weakness of the general gradient-based optimization algorithm and propose to train an LSTM [43] to guide the optimization of the base model. Finn *et al.* [44] further design a unified metalearning framework to search for the best initialization of the model for fast adaptation across tasks. Similarly, Cheng *et al.* [45] integrate the taskwise attention module and the part complementary learning module into the learner for adaptative fast adaptation and robust inductive generalization.

Specific to the field of remote sensing image interpretation, extensive efforts have also been devoted to solving the small data issue. For example, Cheng *et al.* [46] impose a metric learning regularization to increase the discriminability of the model and verify the metric learning techniques can maintain good classification performance in low data regimes. Li *et al.* [47] further propose to organize metric training at the task level to increase the generalization ability of the metric learner across tasks. Li *et al.* [48] utilize an adaptive matcher to automatically seek the optimal measurement of the similarity score between samples. Moreover, Cheng *et al.* [49] and Li *et al.* [50] extend prototypical networks [18] to few-shot object detection task in remote sensing images. Lately, transductive learning schema [51], [52] and knowledge distillation techniques [53] are investigated and exploited to assist the research of few-shot learning.

Unlike the previous works that mainly focus on remote sensing image classification, we tackle the task of few-shot geographic object segmentation, which is more challenging since the interpretation level is raised from image-level classification to pixelwise annotation for more refined image understanding.

### C. Semantic Segmentation With Limited Labeled Samples

There exists a number of prior works focusing on the semantic segmentation of remote sensing images with limited labels. Weakly supervised learning [54], [55] tackles the problem by training image segmenters with rough labels, such as image-level labels or instance-level contours but without pixelwise annotations. Semisupervised learning [56]–[58] considers combining the labeled and unlabeled samples to enhance the learning efficiency in dealing with the conditions where labeled data are scarce. Both the weakly supervised learning and the semisupervised learning schema can significantly reduce the dependence on massive finely annotated samples. Nevertheless, they still require abundant coarsely labeled or unlabeled images for each category to train the model, which may not always be satisfied, especially for rare geographic object categories. Hopefully, transfer learning [59] provides another promising paradigm to solve the low data problem by reusing knowledge learned from other related tasks or domains. The preexisting labeled data for the same task or similar task (also known as the source domain), which may be collected from multisources across various periods, geographic space, and imaging sensors, can be utilized to enhance the current learning task (also known as the target domain) [60], [61]. One limitation of this approach is that the current work mainly focuses on intraclass knowledge migration, which generally assumes the target domain shares the same category space with the source domain, resulting in the poor capacity to handle the segmentation task of novel classes.

Different from the above settings, few-shot semantic segmentation uses very limited annotated samples (1–10) for each novel category, without the need for any other supplementary samples (roughly labeled or unlabeled) of the same category. OSLSM [62] first tackles the challenge of few-shot semantic segmentation by designing a two-branched network architecture to learn an ensemble classifier. SG-One [63] and PANet [16] adopt the idea of prototypical networks to learn robust feature representations for each class and perform segmentation by dense feature comparison. CANet [17] further design an iterative optimization technique to refine the prediction results iteratively. Generally, the existing approaches for few-shot segmentation are affiliated with the metric-based category, which trains the model to learn a static class-agnostic metric space and conducts segmentation in a contrastive way. Despite effectiveness, these methods expose poor compatibility for the geographic objects in remote sensing images as they are initially designed to handle the few-shot segmentation of common objects in natural scenes. The class-specific knowledge is not leveraged, leading to the poor generalization ability for geographic objects with huge intraclass variations. Besides, the traditional metric comparison is conducted on a single scale, which may fail to deal with the scale inconsistency of geographic objects.

## III. Methodology

### A. Task Description

For geospatial object segmentation in remote sensing imagery with only a few labeled samples, we consider the task of few-shot learning. Formally, a $K$-shot segmentation task $T$ is composed of the support set $\mathcal{S}$ and the query set $\mathcal{Q}$. Given $K$ labeled samples of the novel class $c$ from the support set $\mathcal{S}$, the objective is to segment the region of class $c$ from each query image $I_Q$ in the query set $\mathcal{Q}$.

Generally, deep CNNs have difficulty in learning the optimal mapping function in low shot regimes due to the severe imbalance between the huge parameter space and the small data size. One desirable paradigm for the small data dilemma is to borrow the prior knowledge from the existing large-scale dataset (e.g., iSAID [19], [64]). Thus, the entire learning procedure is divided into two phases: metatraining and metatesting. During metatraining, the model has access to an auxiliary dataset that contains adequate labeled samples of the base classes $C_{\text{base}}$. The goal of metatraining is to learn the metaknowledge that can be well generalized for the segmentation tasks of other novel classes. During metatesting, the obtained metamodel is fine-tuned on the small training set and then evaluated by performing few-shot segmentation of novel classes $C_{\text{novel}}$ on the test data. Note that there is no overlap between the base classes and the novel classes ($C_{\text{base}} \cap C_{\text{novel}} = \oslash$).

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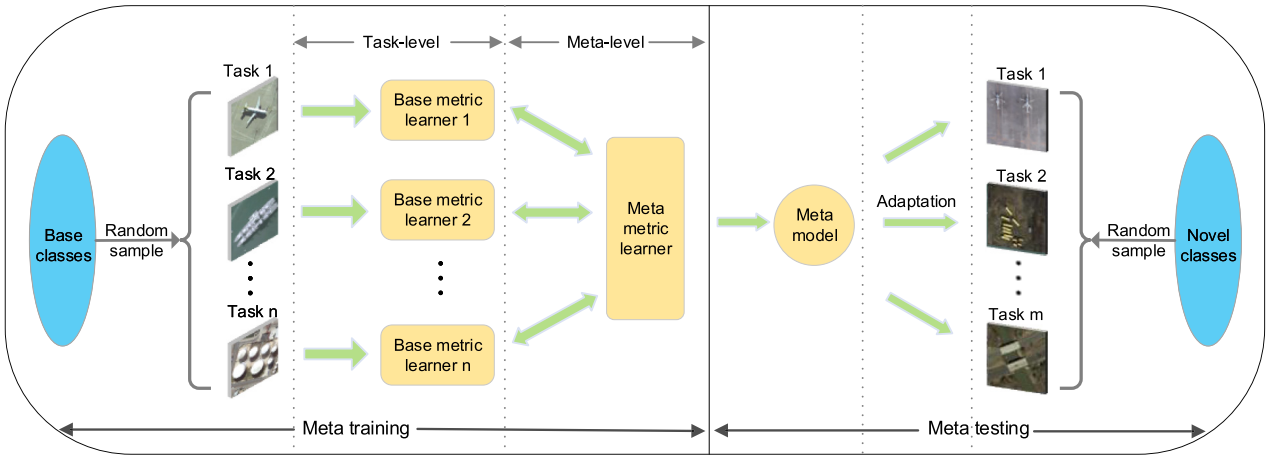



Fig. 2. Overview of the proposed DMML-Net for few-shot geographic object segmentation.

To imitate the few-shot scenario, both metatraining and metatesting are conducted on the constructed episodes. Specifically, for the construction of each episode in metatraining or metatesting, $N$ kinds of geographic object categories are randomly chosen from the base classes $C_{\text{base}}$ or novel classes $C_{\text{novel}}$ as the target classes to be segmented in this episode. Subsequently, samples of the chosen target classes are further randomly selected and organized into a minibatch of tasks $\mathcal{T} = \{T_i | i = 1, 2, \ldots, N\}$ following the definition of the $K$-shot segmentation task. The constructed episode, which consists of $N$ segmentation tasks for $N$ geographic object classes, is treated as the basic cell to be processed for each optimization of DMML-Net. This episode-based learning schema helps improve the generalization ability of the model and enhances the credibility of the evaluation results.

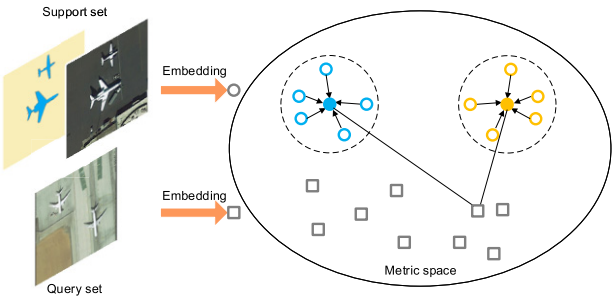



Fig. 3. Illustration for the learning objective of the metric learner in a one-shot example. The metric space is expected to provide interclass separability and intraclass intraclass compactness. The segmentation is transformed into the nearest neighbor classification between the query embeddings and the class-specific prototypes.

### *B. Overview of the Proposed Method*

We use the popular few-shot learning framework for geographic object segmentation in remote sensing imagery, referred to as DMML-Net. The overall goal of DMML-Net is to obtain a metametric model, which is capable of acquiring segmentation skills for target objects from very limited labeled samples, as well as holding flexible scalability to adapt to different learning biases of various segmentation tasks. Fig. 2 illustrates the overview of the proposed framework. The proposed learning framework is composed of a metametric learner and a series of base-metric learners. Both the metametric learner and the base-metric learners adopt the same architectural design of the proposed deep feature pyramid comparison network. The objective of each metric learner is to learn a metric space where the distribution of homogeneous pixels is compact and the distribution of inhomogeneous pixels is alienated, which is shown in Fig. 3. The main difference between the base-metric learner and the metametric learner lies in the source of the learned metric space, where the base-metric learner learns from a single task (task level), while the metalearner learns from a set of tasks (metalevel). After metatraining on the adequate labeled samples of the base classes, the desired metametric model can be obtained by copying values from the metametric learner as parameter initializations and then applied for few-shot segmentation tasks of novel classes with only a gradient steps.

For the organization of the following paragraphs, we first introduce the architecture design of the metric learner for multiscale metric learning and its corresponding inference procedure in one-shot conditions. Then, we expand the scope of the application to the generalized $k$-shot settings ($k > 1$) and introduce the affinity-based fusion mechanism for the deviation calibration of the estimated prototypes. Finally, we come back to the whole of the learning framework and provide a detailed illustration of the metametric training algorithm for the entire network.

### *C. Architecture of the Metric Learner*

To deal with the scale variations of geographic objects, a deep feature pyramid comparison network is proposed as the architecture design of the metric learner to resolve the segmentation tasks in a contrastive manner. In particular, the network extracts class-specific prototypes from the support set for robust representation of intraclass properties. Then, the segmentation process, which can be regarded as pixelwise classification, is conducted via the dense metric comparison

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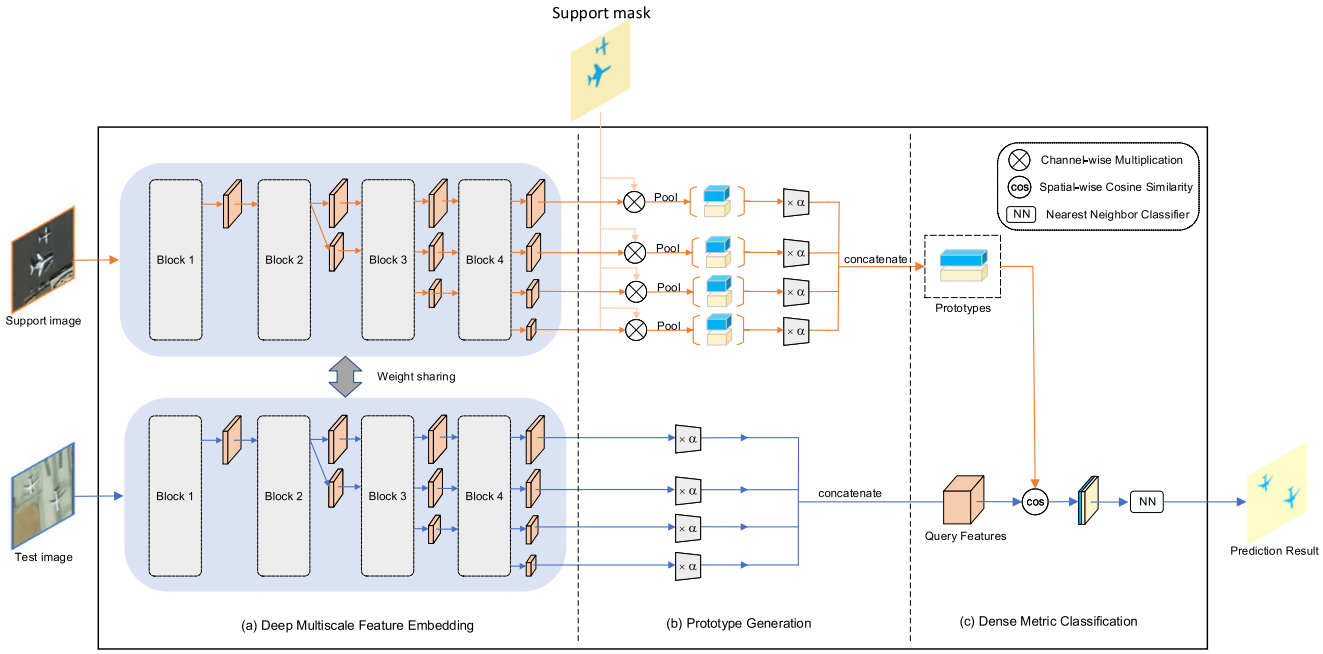



Fig. 4. Inference pipeline of the proposed deep feature pyramid comparison network for multiscale metric learning in a one-shot example. (a) Deep multiscale feature embedding. (b) Prototype generation. (c) Dense metric classification.

at each position of the test image features and the prototypes in the metric space. As illustrated in Fig. 4, the entire inference pipeline for geographic object segmentation consists of three stages: deep multiscale feature embedding, prototype generation, and dense metric classification. We initially begin with the introduction of our network design in one-shot segmentation tasks, and the techniques for the generalized $K$-shot segmentation tasks ($K > 1$) are later introduced in Section III-D.

*1) Deep Multiscale Feature Embedding:* During this process, both the support image and the test image are embedded by the same feature extractor into pyramidal feature representations for the subsequent prototype extraction and spatial metric classification. Different from the ordinary objects in natural scenes generally of consistent size, the geographic objects generally possess large intraclass variations and present with distinct scale sizes in remote sensing imagery. Therefore, a multiscale feature extractor is desired to enrich the intraclass feature descriptors for refined metric comparison.

In this article, we construct the feature extractor based on the HRNet-30 [65]. HRNet is a powerful backbone network for multiscale feature embedding, as well as maintaining high-resolution representations. Specifically, HRNet gradually encodes the input images into multiscale feature representations by the parallel multiresolution convolutions and repeatedly connects them with multiresolution fusions. Here, we keep the original first four blocks for multiscale feature extraction and remove the subsequent operations for feature concatenation and integration in the original design. Given the input image $I$, the feature extractor generates the pyramidal features with four kinds of spatial resolutions. The embedding process can be formulated as

$$I \xrightarrow{f} \mathcal{F} = \left\{ F_i \in \mathbb{R}^{C_i \times H_i \times W_i} | i = 1, 2, 3, 4 \right\} \tag{1}$$

where $f$ denotes the feature extratcor, $\mathcal{F}$ denotes the set of the embedded features for multiple scales, $F_i$ are the embedded features for the $i_{\text{th}}$ scale, $C_i$ is the number of channels for the features at the $i_{\text{th}}$ scale, and $W_i$ and $H_i$ are width and height for the features at the $i_{\text{th}}$ scale. The sizes of the generated pyramidal features correspond to the downsample ratio of 1/4, 1/8, 1/16, and 1/32 to the raw input image $I$.

*2) Propotype Generation:* At this stage, based on the pyramidal feature maps extracted from the support set, we aim to harvest the prototypes for each semantic category, including the background. The class-specific prototypes are 1-D embeddings that are expected to be representative of the intraclass feature abstraction and well-separated for interclass discrimination. To achieve this goal, we design a multiscale prototype extraction technique, which leverages the mask annotation information to extract the subprototypes at each scale and then fuses them to generate the scale-balanced prototypes.

Formally, given a one-shot segmentation task $T$ with support image-mask pair $\mathcal{S} = \{(I_s, M_s)\}$, let $F_{s_i} \in \mathbb{R}^{C_i \times H_i \times W_i}$ be the feature maps generated by the feature extractor at the $i_{\text{th}}$ scale. For the generation of the subprototype $p_{c,i}$ for class $c$ at the $i_{\text{th}}$ scale, we first downsample the mask annotations $M_s$ to the same spatial size of $F_{s_i}$. Then, the resized mask annotations are utilized to filter the noise features that are irrelevant with class $c$ via elementwise multiplication with the corresponding values on the feature map. Subsequently, the subprototype is computed by averaging the spatial values within the valid region of each channel on the filtered features. The entire process can be summarized as

$$p_{c,i} = \frac{\sum_{x,y} F_{s_i}^{(x,y)} \times \mathbb{1}\left[M_{s_i}^{(x,y)} = c\right]}{\sum_{x,y} \mathbb{1}\left[M_{s_i}^{(x,y)} = c\right]} \tag{2}$$

where $M_{s_i}$ denotes the resized mask annotations at the $i_{\text{th}}$ scale. From the perspective of the receptive field theory, the value at

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each spatial location of the feature maps is affected by a particular patch area in the raw input image. The manipulation of extracting subprototypes from the pyramidal feature maps can be understood as statistically averaging the intraclass features with different sliding patch sizes. The extracted subprototypes reflect the intraclass properties of the geographic object at different scales.

Finally, the subprototypes acquired from different scales are rescaled by the scaling coefficients and then concatenated as the final scale-balanced prototypes for the individual support sample. This process can be expressed as

$$p_c = \text{concat}[\alpha_1 p_{c,1}, \alpha_2 p_{c,2}, \alpha_3 p_{c,3}, \alpha_4 p_{c,4}] \tag{3}$$

where $p_c$ represents the scale-balanced prototype of class $c$. The scaling coefficients $\alpha_i$ are a series of learnable scale-specific scalar parameters, acting as the regulator to adjust the proportion of the extracted intraclass properties at each scale.

Similarly, the extracted pyramidal features $\mathcal{F}_t$ for the test image are resized to the same max spatial size in $\mathcal{F}_t$ and then concatenated after rescaling by the same set of scaling coefficients for the subsequent dense metric classification.

*3) Dense Metric Classification:* As the segmentation is equivalent to the pixelwise classification, the network generates the final segmentation results by performing the dense spatial metric classification between the extracted prototypes and the test image feature embeddings.

Given the extracted prototypes $P = \{p_c \in \mathbb{R}^C | c \in \mathcal{C}\}$ for each class $c$ in the label sets $\mathcal{C}$ and the test image features $F_t \in \mathbb{R}^{C\times H\times W}$, we calculate the cosine similarity scores between the prototypes and the feature vector at each spatial location of the test image features. Then, the segmentation probability maps can be obtained via applying the softmax function over the calculated cosine similarity scores across classes. Concretely, the confidence score $\tilde{M}_{t_j}^{(x,y)}$ at location $(x, y)$ of the test image features for class $j$ is calculated as

$$\tilde{M}_{t_j}^{(x,y)} = \frac{\exp\left(\cos\left(p_j, F_t^{(x,y)}\right)\right)}{\sum_{c\in L}\exp\left(\cos\left(p_c, F_t^{(x,y)}\right)\right)} \tag{4}$$

$$\cos(a, b) = \frac{a\cdot b}{\|a\|_2 \cdot \|b\|_2}. \tag{5}$$

The final segmentation mask $\hat{M}$ for label annotation in pixel level is given by the nearest neighbor classification in the upsampled segmentation probability maps

$$\hat{M}_t^{(x,y)} = \arg\max_{c\in\mathcal{L}}\left(\text{Upsample}\left(\tilde{M}_{t_c}^{(x,y)}\right)\right) \tag{6}$$

where the upsampling process is implemented via bilinear interpolation. The generation of the final segmentation mask is equivalent to the aggregation of the metric classification results from multiple scales. The segmentation loss between the obtained segmentation probability score map $\tilde{M}_t$ and ground truth $M_t$ is defined as

$$\mathcal{L}_{(\tilde{M}_t, M_t)} = -\frac{1}{N}\sum_{x,y}\sum_{c\in\mathcal{C}}\mathbb{1}\left[M_t^{(x,y)} = c\right]\cdot\log\left(\tilde{M}_{t_c}^{(x,y)}\right) \tag{7}$$

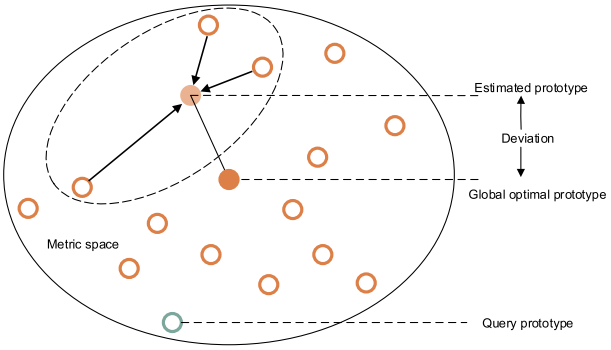



Fig. 5. Schematic of the deviation of the estimated prototypes induced by the intraclass variations.

where $N$ denotes the quantity of the pixels to be classified in the test image.

### *D. Affinity-Based Fusion Mechanism*

To expand the proposed deep feature pyramid comparison network from one-shot learning to the generalized $K$-shot learning ($K > 1$), one basic strategy is to take the average of the individual prototypes. Consequently, the segmentation map can be generated under the guidance of averaged prototypes in the metric space. However, as shown in Fig. 5, the optimal prototypes can be hard to approximate by averaging operation for the lack of the representative object instances. Moreover, considering the characteristic of large intraclass variations that inherently exist on remote sensing imagery, the representativeness of the mean prototypes averaged from the small support set for the query image can be restrictive.

In avoidance of the above issues, we introduce the concept of affinity to measure the intraclass associations between samples. The affinity scores serve as the coefficients of individual prototypes and dynamically reweight them for prototype rectification. As illustrated in Fig. 6, the entire inference process contains a twice metric classification of the query image. The first metric classification is conducted under the guidance of the averaged prototypes and act as the tentative reasoning to generate the coarse segmentation mask for the query image. The second metric classification is conducted under the guidance of the rectified prototypes and generates the final refined segmentation mask.

For the generation of the rectified prototypes, the learner first extracts the pseudoprototypes $\tilde{\mathcal{P}}_Q = \{\tilde{q}_c | c \in \mathcal{C}\}$ for the query sample based on the extracted query features and corresponding coarse segmentation mask. Subsequently, the affinity between the support and query samples is calculated. Specifically, the affinity score is estimated by the exponential harmonization of the cosine similarity between the prototypes. The concrete process can be expressed as

$$w_{c,k} = \frac{\exp\left(\cos\left(p_{c,k}, \tilde{q}_c\right)\right)}{\sum_{k=1}^{K}\exp\left(\cos\left(p_{c,k}, \tilde{q}_c\right)\right)} \tag{8}$$

where $w_{c,k}$ denotes the affinity score on class $c$ between the $k_{\text{th}}$ support sample and the query sample, and $p_{c,k}$ is the

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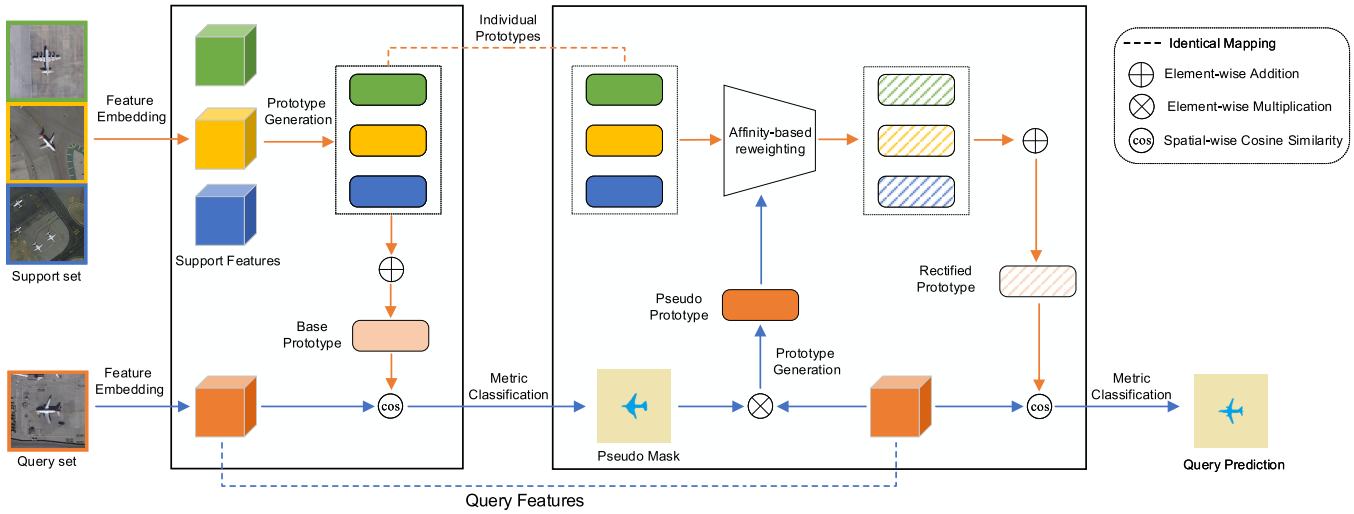



Fig. 6. Illustration of the affinity-based fusion mechanism in a three-shot example. The entire inference process contains a twice metric classification of the query image (see Section III-D for details).

prototype for class $c$ extracted from the $k_{\text{th}}$ support sample. The obtained affinity score reflects the representativeness of the individual support prototype for the congeneric pixels in the query image. Then, the computed affinity scores act as the coefficients for the reweighting of each inidividual prototype during prototype rectification. The rectified prototype $\hat{p}_c$ for class $c$ is calculated as

$$\hat{p}_c = \sum_{k=1}^{K} w_{c,k} \cdot p_{c,k}. \tag{9}$$

In addition, to increase the fidelity of the query pseudoprototypes, we add a regularization term on the query pseudomasks generated by the first metric classification. The final loss $L_{\text{seg}}$ to be optimized can be expressed as

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{(\tilde{M}_{q_2}, M_q)} + \lambda \mathcal{L}_{(\tilde{M}_{q_1}, M_q)} \tag{10}$$

where $M_q$ denotes the ground-truth annotation for the query image, $\tilde{M}_{q_1}$ is the probability map for query pseudoprediction generated by the first metric classification, $\tilde{M}_{q_2}$ represents the probability map for the final refined query prediction generated by the second metric classification, and $\lambda$ serves as factor to control the regularization strength.

### E. Metametric Training Algorithm

Considering the impact of the large interclass distribution discrepancy, the traditional metric training strategy exhibits low learning efficiency due to the poor transferability of the profitable feature representations across classes. Besides, the peculiar metric learning requirement of each task is neglected for the lack of task-specific metric adaptation. Based on this analysis, we introduce the metametric training algorithm. The goal is to accumulate the metametric knowledge by learning good parameter initializations and enhance the suitability of the metamodel for the individual task by the task-specific adaptation strategy.

The entire metametric training algorithm is developed based on the episode-based learning schema. Given the metametric learner $f_\theta$ and a constructed episode $\mathcal{T} = \{T_i | i = 1, 2, \ldots, N\}$, a series of base metric learners $f_{\theta_i}$ are created via the instantiations of the metametric learner $f_\theta$ for each task $T_i$. The optimization process within an episode is explicitly decoupled into two stages, including task-specific knowledge adaptation and cross-task knowledge accumulation.

During the process of task-specific knowledge adaptation, metric learning is conducted at the task level to mimic the fast adaptation process for novel tasks. Specifically, for each task $T_i$, the corresponding base metric learner $f_{\theta_i}$ is first required to extract prototypes from the support set $S_i$ and utilize the obtained prototype representations to generate the predictions for the support images. Then, the parameters $\theta_i$ of base metric learner $f_{\theta_i}$ are updated via the standard gradient optimization

$$\theta_i' = \theta_i - \alpha \nabla_{\theta_i} \mathcal{L}_{S_i} \tag{11}$$

where $\alpha$ stands for the learning rate of the base learners and $\mathcal{L}_{S_i}$ denotes the loss for the generated predictions on the support set $S_i$. The updated weights of the metric learner are expected to cater to the task-specific metric learning bias, including which kind of feature representations should be strengthened and which kinds of feature representations should be weakened.

During the process of cross-task knowledge accumulation, metric learning is conducted at the metalevel to gradually accumulate cross-task knowledge. Concretely, the optimized base metric learners $f_{\theta_i'}$ are utilized to make predictions for the query images of each task. Then, the corresponding gradients for the parameters of each base metric learner are calculated and fed back to the metametric learner. To integrate the learning results of base metric learners from multiples tasks, the parameters $\theta$ of the metametric learner $f_\theta$ are optimized as

$$\theta' = \theta - \beta \sum_{i=1}^{n} \nabla_{\theta_i} \mathcal{L}_{Q_i} \tag{12}$$

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**Algorithm 1** Metametric Learning for Few-Shot Geographic Object Segmentation

**Require:** Meta-train set $D_{\text{meta-train}}$ which is constructed by repeatedly random sampling episodes from the base classes $C_{\text{base}}$, meta-test set $D_{\text{meta-test}}$ which is constructed by repeatedly random sampling episodes from the novel classes $C_{\text{novel}}$, learning rate $\alpha$ for base-metric learners, learning rate $\beta$ for meta-metric learners, number of tasks $n$ in an episode $\mathcal{T}$
**Ensure:** Parameters $\theta$ of the meta-metric learner $f_\theta$, Parameters $\theta_i$ of the $i_{\text{th}}$ base-metric model $f_{\theta^i}$
****** Meta training ******
Initialize $\theta$
**for** *each episode* $\mathcal{T} \in D_{meta\text{-}train}$ **do**
  **for** *each task* $T_i = (S_i, Q_i) \in \mathcal{T}$ **do**
    Initialize $\theta_i \leftarrow \theta$
    Use $f_{\theta_i}$ to generate predictions for the support set $S_i$ following Eqns. (1) - (6)
    Compute the loss $\mathcal{L}_{S_i}$ on the predictions of the support set $S_i$ following Eqns. (7)
    Update $\theta_i$ for task-specific knowledge adaptation: $\theta_i' \leftarrow \theta_i - \alpha \nabla_{\theta_i} \mathcal{L}_{S_i}$
    Use $f_{\theta_i'}$ to generate predictions for the query set $Q_i$ following Eqns. (1) - (6) and Eqns. (8) - (9)
    Compute the loss $\mathcal{L}_{Q_i}$ on the predictions of the query set $Q_i$ following Eqns. (7) and Eqns. (10)
    Compute the gradient update $\nabla_i$ of $\theta_i$: $\nabla_i = \nabla_{\theta_i} \mathcal{L}_{Q_i}$
  **end**
  Update $\theta$ for cross-task knowledge accumulation: $\theta' \leftarrow \theta - \beta \sum_i \nabla_i$
**end**
**return** $\theta$
****** Meta testing ******
**for** *each episode* $\mathcal{T} \in D_{meta\text{-}test}$ **do**
  **for** *each task* $T_i = (S_i, Q_i) \in \mathcal{T}$ **do**
    Initialize $\theta_i \leftarrow \theta$
    Use $f_{\theta_i}$ to generate predictions for the support set $S_i$ following Eqns. (1) - (6)
    Compute the loss $\mathcal{L}_{S_i}$ on the predictions of the support set $S_i$ following Eqns. (7)
    Update $\theta_i$ for task-specific knowledge adaptation: $\theta_i' \leftarrow \theta_i - \alpha \nabla_{\theta_i} \mathcal{L}_{S_i}$
    Use $f_{\theta_i'}$ to generate predictions for the query set $Q_i$ following Eqns. (1) - (6) and Eqns. (8) - (9)
  **end**
**end**

where $\beta$ is the learning rate of the metametric learner and $\mathcal{L}_{Q_i}$ is the loss for the generated predictions on the query set $Q_i$. Following this way, the desired cross-task knowledge can be gradually collected and saved in the metametric learner. Here, the cross-task knowledge can be comprehended as the model parameters that are capable of generating large gains for the majority of tasks after minor changes.

The entire pipeline for the metametric learning algorithm is summarized in Algorithm 1. After the metatraining under the proposed metametric learning paradigm, the metamodel can be generated by the instantiation of the metametric learner and applied for the various few-shot segmentation tasks after only a few gradient steps.

## IV. Experiment and Analysis

In this section, we perform experiments to evaluate the effectiveness of the proposed method.

### A. Description of Datasets

In the experiments, we evaluate the proposed DMML-Net on two remote sensing image datasets designed for few-shot geographic object segmentation.

*1) Mini-iSAID Dataset:* We introduce a challenging benchmark dataset, named mini-iSAID, for few-shot geographic object segmentation on remote sensing images based on the existing public available dataset: Instance Segmentation in Aerial Images Dataset (iSAID). iSAID is a large-scale dataset for evaluating instance segmentation and semantic segmentation algorithms, consisting of 2806 high-resolution remote sensing images from multiple imaging sensors and platforms. The original dataset contains 15 kinds of typical geographic objects: baseball diamond, bridge, ground track field, harbor, helicopter, large vehicle, plane, roundabout, ship, small vehicle, soccer ball field, storage tank, swimming pool, and tennis court. However, this dataset exposes some maladjustments for evaluating few-shot segmentation algorithms, such as the rigorous computation requirements on the engineering resources, the extremely large foreground-background area ratio that dilutes the evaluation metrics, and the scarcity of object instances for rare classes. Thus, we make some modifications in the original dataset to enhance its compatibility for few-shot segmentation experiments, such as selecting regions of interest and cropping them into patches, enlarging the number of instance samples for rare classes, and correcting the mislabeled annotations. Besides, the images and annotations for other two kinds of typical geographic objects, including water and building, are introduced to enlarge the original dataset. Finally, the enlarged dataset contains instances and corresponding annotations for 17 kinds of geographic objects, and the resolution of all the images is fixed to be 513 $\times$ 513 pixels for fits in memory and fair evaluation. The statistical results on the quantity distribution of labeled samples across classes for mini-iSAID can be seen in Table I.

*2) DLRSD Dataset:* The Dense Labeling Remote Sensing Dataset (DLRSD) is a publicly available dataset for evaluating multilabel remote sensing image retrieval and semantic segmentation algorithms. DLRSD contains 2100 RGB images in total, and the image sizes are fixed as 256 $\times$ 256 pixel. The following 17 object classes, i.e., airplane, bare soil, building, car, chaparral, court, dock, field, grass, mobile home, pavement, sand, sea, ship, tank, tree, and water, are selected as the target geographic object category to be segmented. The distribution fn the quantity of labeled samples across classes for DLRSD is listed in Table II.

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TABLE I
NUMBER OF IMAGES PER CLASS FOR THE MINI-iSAID DATASET

| Split ID | Category | #Num of imgs in train set | #Num of imgs in val set |
|---|---|---|---|
| A | Baseball diamond | 396 | 128 |
| | Basketball court | 406 | 134 |
| | Bridge | 436 | 152 |
| | Building | 800 | 106 |
| B | Ground track field | 490 | 162 |
| | Harbor | 1328 | 462 |
| | Helicopter | 180 | 44 |
| | Large vehicle | 1336 | 314 |
| C | Plane | 2054 | 442 |
| | Roundabout | 422 | 104 |
| | Ship | 848 | 314 |
| | Small vehicle | 1292 | 296 |
| D | Soccer ball field | 552 | 172 |
| | Storage tank | 634 | 162 |
| | Swimming pool | 652 | 120 |
| | Tennis court | 814 | 176 |
| | Water | 1000 | 102 |

TABLE II
NUMBER OF IMAGES PER CLASS FOR THE DLRSD DATASET

| Split ID | Category | #Num of imgs in train set | #Num of imgs in val set |
|---|---|---|---|
| A | Airplane | 140 | 30 |
| | Bare soil | 1022 | 243 |
| | Building | 996 | 215 |
| | Cars | 1258 | 268 |
| B | Chaparral | 164 | 34 |
| | Court | 150 | 30 |
| | Dock | 140 | 30 |
| | Field | 138 | 34 |
| C | Grass | 1380 | 287 |
| | Mobile home | 142 | 31 |
| | Pavement | 1858 | 402 |
| | Sand | 418 | 82 |
| D | Sea | 140 | 31 |
| | Ship | 142 | 32 |
| | Tank | 140 | 30 |
| | Tree | 1438 | 302 |
| | Water | 294 | 61 |

### B. Implementation Details

We implement our model in the PyTorch framework, and GPU is NVIDIA Tesla P100. Our backbone networks are pretrained on ImageNet [66] and then fine-tuned on NWPU-RESISC45 [67] to obtain better generalization performance for remote sensing image interpretation. During metatraining, the models are trained end-to-end by SGD with the momentum of 0.9 for 40 000 episodes. The number of tasks or metric learners within an episode is set as 2, and the weight decay is 0.0005. The learning rate for both the base-metric learners and the metametric learner is initialized as 0.001 and reduced by 0.1 every 10 000 episodes. As for the selection of the hyperparameter $\lambda$ on the loss function in $k$-shot settings ($k > 1$), we set it as 0.4. The detailed discussion on the impact of the hyperparameter $\lambda$ is later shown in Section IV-E2. During metatesting, the maximum gradient step is set as 5 for one- and five-shot segmentation tasks and 10 for ten-shot segmentation tasks with the step size of 0.01. Note that all the test results are obtained without any additional postprocessing (for example, multiscale testing, and dense conditional random field).

### C. Evaluation Metrics

We adopt two metrics to evaluate the performance of few-shot segmentation algorithms, including mean-F score and foreground-background F score (FB-F score). We first introduce the primary metrics concepts, including TP, FP, FN, precision, and recall. Then, we illustrate the adopted evaluation criteria in detail.

*1) TP, FP, and FN:* TP is the abbreviation for true positive, indicating the number of correctly classified pixels for the target class. FP stands for false positive and corresponds to the number of pixels that actually belong to other categories yet are wrongly classified into the target category. Similarly, FN represents false negative and refers to the number of pixels that actually belong to the target category yet are wrongly divided into other classes.

*2) Precision and Recall:* Precision is defined as the fraction of true positives among all the positives for the target class predicted by the classifier. The recall is defined as the ratio of true positives and all the pixels that should be classified as positive for the target category. These two metrics are calculated as

$$\text{Precision} = \frac{\text{TP}}{(\text{TP} + \text{TN})} \tag{13}$$

$$\text{Recall} = \frac{\text{TP}}{(\text{TP} + \text{FN})}. \tag{14}$$

*3) F Score, Mean-F Score, and FB-F Score:* F score is defined as the harmonic mean of precision and recall for the target category

$$\text{F} = 2 \times \frac{\text{Precision} \times \text{Recall}}{(\text{Precision} + \text{Recall})}. \tag{15}$$

Specific to the field of few-shot segmentation, the evaluation process generally consists of a series of binary foreground-background segmentation tasks, and the F score is calculated from the sampled binary segmentation tasks of each category. The mean-F score is defined as the statistical average of the obtained class-specific F score for all foreground object classes. FB-F score treats all foreground objects as one foreground category and averages the foreground F score and background F score.

In comparison, the mean-F score metric considers the differences between different foreground object classes and can more accurately reflect the few-shot segmentation abilities across classes. In contrast, the FB-F score metric tries to evaluate the performance of the model from a more general respective and take the predictions for the background category into the calculation. Sometimes, the FB-F score metric can lead to biased evaluation results toward the background category with a larger pixel ratio. Thus, we adopt the mean-F

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TABLE III

FEW-SHOT SEGMENTATION PERFORMANCE (MEAN-F SCORE, %) ON THE MINI-iSAID DATASET. WE CONDUCT FOUR-FOLDER CROSS VALIDATION UNDER THE DIFFERENT SPLIT SETTINGS OF THE BASE CLASSES AND NOVEL CLASSES. THE ABBREVIATIONS OF THE NAMES ARE DEFINED AS FOLLOWS: C1: BASEBALL DIAMOND, C2: BASKETBALL COURT, C3: BRIDGE, C4: BUILDING, C5: GROUND TRACK FIELD, C6: HARBOR, C7: HELICOPTER, C8: LARGE VEHICLE, C9: PLANE, C10: ROUNDABOUT, C11: SHIP, C12: SMALL VEHICLE, C13: SOCCER BALL FIELD, C14: STORAGE TANK, C15: SWIMMING POOL, C16: TENNIS COURT, AND C17: WATER

| Shot | Method | Split-A | | | | | Split-B | | | | | Split-C | | | | | Split-D | | | | | | Mean |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | c1 | c2 | c3 | c4 | **mean** | c5 | c6 | c7 | c8 | **mean** | c9 | c10 | c11 | c12 | **mean** | c13 | c14 | c15 | c16 | c17 | **mean** | |
| 1-shot | SG-One | 41.6 | 51.7 | 49.7 | 49.2 | 48.1 | 28.0 | 21.1 | 19.0 | 41.3 | 27.4 | 42.4 | 36.7 | 33.0 | 29.7 | 35.5 | 43.8 | 68.4 | 28.7 | 44.8 | 60.1 | 49.2 | 40.0 |
| | CANet | 40.0 | 44.4 | 34.7 | 42.7 | 40.4 | 20.8 | 15.4 | 23.8 | 29.1 | 22.3 | 33.8 | 37.2 | 33.3 | 20.0 | 31.1 | 38.0 | 67.8 | 17.6 | 36.9 | 50.0 | 42.1 | 34.0 |
| | PANet | 50.4 | 61.3 | 39.6 | 60.9 | 53.1 | 33.6 | 34.6 | 20.1 | 48.3 | 34.1 | 51.2 | 45.4 | 39.5 | 34.1 | 42.5 | 57.4 | 71.7 | 23.3 | 53.0 | 72.0 | 55.5 | 46.3 |
| | PANet+ | 58.7 | 68.6 | 47.7 | 61.3 | 59.1 | 39.2 | 34.0 | 26.1 | 57.7 | 41.8 | 59.7 | 50.6 | 57.1 | 51.5 | 54.8 | 62.4 | 76.9 | 25.0 | 63.3 | 66.5 | 58.8 | 53.6 |
| | DMML-Net | 63.9 | 63.1 | 58.0 | 63.2 | **62.1** | 42.1 | 49.5 | 36.2 | 62.5 | **47.6** | 66.0 | 72.1 | 60.3 | 58.7 | **64.3** | 73.5 | 78.1 | 29.2 | 67.6 | 71.8 | **64.1** | **59.5** |
| 5-shot | SG-One | 45.2 | 57.6 | 54.5 | 53.9 | 52.8 | 29.5 | 31.1 | 21.7 | 46.0 | 32.1 | 43.8 | 43.1 | 40.3 | 37.7 | 41.2 | 45.8 | 71.2 | 30.6 | 47.7 | 64.2 | 51.9 | 44.5 |
| | CANet | 45.3 | 51.1 | 34.1 | 46.1 | 44.1 | 27.7 | 20.3 | 31.3 | 37.6 | 29.2 | 36.6 | 44.7 | 36.6 | 22.9 | 35.2 | 42.7 | 69.9 | 19.4 | 38.5 | 56.8 | 45.5 | 38.5 |
| | PANet | 59.4 | 64.8 | 42.8 | 65.0 | 58.0 | 31.3 | 39.3 | 27.5 | 54.7 | 38.2 | 58.6 | 55.5 | 49.7 | 47.2 | 52.8 | 56.2 | 81.0 | 30.2 | 66.5 | 79.4 | 62.7 | 52.9 |
| | PANet+ | 58.1 | 73.2 | 49.4 | 67.3 | 62.0 | 37.6 | 36.2 | 35.6 | 62.3 | 43.9 | 61.9 | 51.1 | 61.6 | 54.9 | 58.6 | 62.5 | 78.4 | 23.5 | 66.2 | 85.4 | 63.2 | 56.9 |
| | DMML-Net | 70.9 | 80.5 | 66.6 | 73.1 | **72.8** | 53.1 | 58.5 | 46.0 | 70.4 | **57.0** | 72.6 | 79.4 | 71.2 | 66.0 | **72.3** | 82.8 | 85.1 | 47.3 | 80.2 | 89.7 | **77.0** | **69.8** |
| 10-shot | SG-One | 47.7 | 59.8 | 58.2 | 56.3 | 55.5 | 31.1 | 34.0 | 27.7 | 49.4 | 35.6 | 46.1 | 47.8 | 41.8 | 42.8 | 44.6 | 48.2 | 74.6 | 33.9 | 49.5 | 69.2 | 55.1 | 47.7 |
| | CANet | 47.9 | 54.1 | 40.4 | 47.3 | 47.4 | 26.6 | 27.3 | 30.6 | 39.9 | 31.1 | 38.0 | 46.3 | 39.5 | 27.1 | 37.8 | 53.0 | 70.7 | 21.3 | 40.7 | 64.7 | 50.1 | 44.8 |
| | PANet | 59.0 | 69.3 | 46.0 | 65.6 | 60.0 | 32.6 | 39.8 | 28.6 | 55.3 | 39.1 | 59.2 | 56.5 | 55.4 | 47.7 | 54.7 | 56.6 | 81.4 | 32.0 | 65.9 | 78.6 | 62.9 | 54.2 |
| | PANet+ | 53.3 | 74.1 | 55.8 | 70.2 | 63.3 | 36.3 | 38.5 | 37.9 | 63.6 | 44.1 | 67.0 | 53.1 | 62.9 | 56.2 | 59.3 | 65.8 | 80.7 | 23.6 | 67.3 | 87.4 | 65.0 | 57.9 |
| | DMML-Net | 75.1 | 82.2 | 68.7 | 75.9 | **75.5** | 71.9 | 62.4 | 47.7 | 71.5 | **63.4** | 76.3 | 83.0 | 75.3 | 71.5 | **76.5** | 87.5 | 87.2 | 61.8 | 86.2 | 93.5 | **83.2** | **74.6** |

TABLE IV

FEW-SHOT SEGMENTATION PERFORMANCE (FB-F SCORE, %) ON THE MINI-iSAID DATASET. WE CONDUCT FOUR-FOLDER CROSS VALIDATION UNDER THE DIFFERENT SPLIT SETTINGS OF THE BASE CLASSES AND NOVEL CLASSES

| Method | Backbone | 1-shot | | | | | 5-shot | | | | | 10-shot | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Split-A | Split-B | Split-C | Split-D | **mean** | Split-A | Split-B | Split-C | Split-D | **mean** | Split-A | Split-B | Split-C | Split-D | **mean** |
| SG-One | VGG-16 | 73.2 | 62.8 | 65.9 | 74.9 | 69.2 | 75.9 | 65.7 | 68.7 | 76.1 | 71.6 | 76.4 | 66.6 | 70.1 | 78.7 | 73.0 |
| CANet | ResNet-50 | 68.5 | 60.6 | 63.1 | 70.3 | 65.6 | 70.9 | 63.7 | 65.7 | 72.6 | 68.2 | 71.7 | 65.2 | 67.4 | 74.9 | 69.8 |
| PANet | VGG-16 | 76.8 | 66.9 | 68.9 | 79.3 | 73.0 | 79.1 | 68.5 | 74.2 | 82.8 | 76.2 | 80.1 | 68.6 | 75.3 | 82.6 | 76.6 |
| PANet+ | HRNet-30 | 78.5 | 70.4 | 76.2 | 79.6 | 76.2 | 80.0 | 71.2 | 77.9 | 83.9 | 78.3 | 81.1 | 70.9 | 78.5 | 85.1 | 78.9 |
| DMML-Net | HRNet-30 | 79.8 | 73.6 | 80.3 | 83.5 | **79.3** | 85.8 | 78.5 | 84.6 | 91.3 | **85.0** | 88.4 | 84.9 | 86.9 | 94.0 | **88.5** |

score metric as the major evaluation metric and the FB-F score metric as the auxiliary metric in our experiments.

### *D. Experimental Results and Comparisons*

We compare the proposed DMML-Net method with several prevalent few-shot segmentation methods in computer version, including SG-One [63], CANet [17], and PANet [16]. Besides, for fair comparisons, the backbone of PANet is further upgraded from the VGG-16 [68] to the HRNet-30 and denoted as PANet+. A series of comparison experiments are conducted on the typical settings in few-shot geographic object segmentation, i.e., one-, five-, and ten-shot segmentations. Note that the affinity-based fusion mechanism is not utilized for DMML-Net in one-shot conditions as this technique only takes effect under the conditions that multiple support samples exist.

The entire evaluation process is conducted in a cross-validation manner. Generally, the 17 kinds of geographic objects in mini-iSAID or DLRSD are randomly divided into four splits in support of the $k$-folder cross validation ($k = 4$), as shown in Tables I and II. For the experiment of each folder, object classes of three splits are selected as the base classes for the metatraining of the networks. The object classes of the left split are chosen as the novel classes to measure the few-segmentation abilities of the trained models. Specifically, we first take the average of the mean-F score and FB-F score from three trials with different random seeds, each trial containing 1000 episodes. The final evaluation results for the specific metrics are computed by averaging the statistical results from each folder.

For the experiments on the mini-iSAID dataset, Tables III and IV and Fig. 7 show the performance of the proposed DMML-Net and the comparison methods under different numbers of shots (i.e., the quantity of the support samples for each novel category). As shown in Table III, our proposed DMML-Net gains superior performance than all the comparison methods. More specifically, in comparison with the current state-of-art few-shot segmentation algorithm PANet+, DMML-Net obtains a mean-F score 5.9% higher in the one-shot setting, 12.9% higher in the five-shot setting, and 16.7% higher in the ten-shot setting. In terms of the evaluation for the class-specific segmentation accuracy, DMML-Net still achieves obvious performance improvement, especially for the hard segmentation tasks (i.e., c5-ground track field, c6-harbor, c7-helicopter, and c15-swimming pool). Here, hard segmentation tasks refer to the geographic objects

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Fig. 7. Qualitative segmentation results of our model on the mini-iSAID dataset. Images of the same column correspond to the segmentation task of the specific geographic object category. The concrete correspondence relations are (a) baseball diamond, (b) building, (c) plane, (d) soccer ball field, (e) storage tank, (f) tennis court, and (g) water.

with large intraclass variations and poor adaptability to the metric knowledge borrowed from other classes. In traditional metric learning methods, the segmentation tasks for hard geographic objects generally expose poor accuracy. Unlike the traditional metric-based few-shot segmentation methods that exhibit jagged performance over the segmentation tasks of different classes, DMML-Net attains relatively well-balanced segmentation results for all the classes. This phenomenon demonstrates the superiority of DMML-Net on the general suitability for the few-shot segmentation tasks of novel categories.

Besides, another important fact can be observed from Tables III and IV is that the mean segmentation performance of each method all gains significant improvement with the increase in the number of support samples. However, from the perspective of the class-specific evaluation, the segmentation models, sometimes, attain performance degradation slightly with the increase in the shots for some geographic objects. For example, the F score for the baseball diamond category (c1) of the ten-shot setting drops by 4.8% compared with the score of the five-shot experiment in PANet+. The F score for the ground track field (c5) of the five-shot setting is 2.3% lower than the score of the one-shot experiment in PANet. We conjecture this phenomenon from two aspects. On the one hand, the target models to be evaluated for $k$-shot learning are obtained via the $k$-shot task-based episodic training. Hence, the evaluation results simultaneously suffer from the impact of the model and the volume of the support set. On the other hand, the prototypes estimated from the small training set can be easily disturbed by the individual noise samples. Specifically, the average fusion strategy neglects the quality of individual support samples and, sometimes, dilutes the valid feature representations. The increase in the number of the support samples is beneficial for the mean segmentation accuracy across classes yet may be harmful to some individual categories.

In addition, a series of experiments are conducted on another benchmark dataset, DLRSD, to further evaluate the performance of DMML-Net and other comparison algorithms. The corresponding experimental results on the DLRSD dataset are listed in Tables V and VI and reflect that the proposed DMML-Net still maintains the highest segmentation accuracy. More concretely, compared with the PANet+, DMML-Net achieves a mean-F score 2.9% higher in the one-shot setting,

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TABLE V

FEW-SHOT SEGMENTATION PERFORMANCE (MEAN-F SCORE, %) ON THE DLRSD DATASET. WE CONDUCT FOUR-FOLDER CROSS VALIDATION UNDER THE DIFFERENT SPLIT SETTINGS OF THE BASE CLASSES AND NOVEL CLASSES. THE ABBREVIATIONS OF THE NAMES ARE DEFINED AS FOLLOWS: C1: AIRPLANE, C2: BARE SOIL, C3: BUILDINGS, C4: CARS, C5: CHAPARRAL, C6: COURT, C7: DOCK, C8: FIELD, C9: GRASS, C10: MOBILE HOME, C11: PAVEMENT, C12: SAND, C13: SEA, C14: SHIP, C15: TANKS, C16: TREES, AND C17: WATER

| Shot | Method | Split-A | | | | | Split-B | | | | | Split-C | | | | | Split-D | | | | | | Mean |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | c1 | c2 | c3 | c4 | **mean** | c5 | c6 | c7 | c8 | **mean** | c9 | c10 | c11 | c12 | **mean** | c13 | c14 | c15 | c16 | c17 | **mean** | |
| 1-shot | SG-One | 18.3 | 36.3 | 37.3 | 11.1 | 25.7 | 51.3 | 21.1 | 18.5 | 89.2 | 45.0 | 33.8 | 44.5 | 33.5 | 30.1 | 35.5 | 31.2 | 33.9 | 24.8 | 23.9 | 38.7 | 30.5 | 34.2 |
| | CANet | 16.1 | 37.5 | 27.4 | 10.2 | 22.8 | 39.7 | 17.6 | 22.5 | 78.9 | 39.7 | 31.3 | 42.3 | 29.7 | 28.1 | 32.9 | 41.5 | 28.7 | 24.6 | 19.6 | 35.1 | 29.9 | 31.3 |
| | PANet | 23.6 | 42.1 | 49.1 | 16.2 | 32.8 | 41.3 | 51.3 | 33.2 | 91.1 | 54.2 | 43.7 | 50.8 | 50.0 | 42.0 | 46.6 | 80.0 | 53.3 | 38.9 | 44.5 | 62.0 | 55.7 | 47.3 |
| | PANet+ | 49.4 | 44.5 | 57.3 | 25.0 | 44.0 | 67.6 | 68.4 | 64.2 | 91.9 | 73.0 | 46.6 | 75.5 | 54.9 | 52.4 | 57.3 | 87.6 | 75.3 | 44.9 | 49.1 | 67.8 | 65.0 | 59.8 |
| | DMML-Net | 58.1 | 47.5 | 59.6 | 32.6 | **49.4** | 70.0 | 63.8 | 68.9 | 92.5 | **73.8** | 48.6 | 75.2 | 56.3 | 54.9 | **58.8** | 91.5 | 78.8 | 53.1 | 47.8 | 72.4 | **68.7** | **62.7** |
| 5-shot | SG-One | 22.4 | 40.4 | 52.1 | 15.3 | 32.6 | 54.5 | 33.1 | 21.5 | 93.1 | 50.5 | 34.9 | 60.4 | 36.0 | 39.3 | 42.7 | 38.1 | 35.1 | 27.1 | 24.1 | 47.5 | 34.4 | 40.0 |
| | CANet | 21.7 | 41.5 | 35.6 | 13.1 | 28.0 | 43.2 | 22.3 | 26.2 | 82.1 | 43.5 | 34.5 | 46.9 | 32.7 | 31.2 | 36.3 | 49.3 | 31.2 | 28.3 | 21.1 | 41.6 | 34.3 | 35.5 |
| | PANet | 34.3 | 44.7 | 60.8 | 21.6 | 40.4 | 54.1 | 62.5 | 39.4 | 94.1 | 62.5 | 48.6 | 57.6 | 52.1 | 53.8 | 53.0 | 81.4 | 58.5 | 45.1 | 46.3 | 65.5 | 59.4 | 53.8 |
| | PANet+ | 58.8 | 44.4 | 63.6 | 36.3 | 50.8 | 72.8 | 74.2 | 65.2 | 95.4 | 76.9 | 45.7 | 76.4 | 59.6 | 64.5 | 61.5 | 93.8 | 77.6 | 61.6 | 51.2 | 76.9 | 72.2 | 65.4 |
| | DMML-Net | 63.6 | 53.8 | 68.5 | 49.0 | **58.7** | 75.2 | 79.1 | 73.0 | 94.9 | **80.6** | 56.4 | 76.4 | 64.6 | 74.4 | **68.0** | 95.4 | 82.0 | 71.9 | 66.3 | 86.8 | **80.5** | **71.9** |
| 10-shot | SG-One | 27.4 | 42.5 | 61.9 | 18.7 | 37.6 | 55.7 | 36.1 | 25.0 | 94.1 | 52.7 | 46.4 | 61.4 | 42.2 | 41.5 | 47.9 | 49.4 | 44.2 | 34.5 | 28.2 | 47.2 | 40.7 | 44.7 |
| | CANet | 25.2 | 43.6 | 44.8 | 15.9 | 32.4 | 46.3 | 27.8 | 29.1 | 84.4 | 46.9 | 38.6 | 51.3 | 36.4 | 35.9 | 40.6 | 55.7 | 37.9 | 35.7 | 24.9 | 44.3 | 39.7 | 39.9 |
| | PANet | 40.5 | 50.1 | 62.8 | 23.8 | 44.3 | 55.5 | 67.0 | 43.3 | 96.0 | 65.4 | 51.8 | 64.9 | 59.4 | 61.5 | 59.4 | 82.9 | 61.9 | 49.6 | 49.6 | 68.3 | 62.5 | 57.9 |
| | PANet+ | 55.4 | 52.7 | 68.7 | 35.4 | 53.0 | 69.1 | 79.6 | 69.3 | 94.5 | 78.1 | 50.0 | 78.0 | 60.1 | 68.3 | 64.1 | 94.0 | 82.5 | 65.5 | 60.4 | 84.4 | 77.3 | 68.1 |
| | DMML-Net | 72.4 | 60.3 | 74.7 | 66.0 | **68.3** | 76.8 | 86.4 | 76.1 | 97.0 | **84.1** | 61.4 | 77.9 | 73.3 | 78.0 | **72.6** | 94.7 | 84.6 | 82.5 | 73.4 | 89.8 | **85.0** | **77.5** |

TABLE VI

FEW-SHOT SEGMENTATION PERFORMANCE (FB-F SCORE, %) ON THE DLRSD DATASET. WE CONDUCT FOUR-FOLDER CROSS VALIDATION UNDER THE DIFFERENT SPLIT SETTINGS OF THE BASE CLASSES AND NOVEL CLASSES

| Method | Backbone | 1-shot | | | | | 5-shot | | | | | 10-shot | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Split-A | Split-B | Split-C | Split-D | **mean** | Split-A | Split-B | Split-C | Split-D | **mean** | Split-A | Split-B | Split-C | Split-D | **mean** |
| SG-One | VGG-16 | 53.3 | 63.5 | 55.9 | 54.2 | 56.7 | 58.0 | 66.8 | 58.3 | 55.3 | 59.6 | 60.7 | 68.7 | 62.2 | 60.9 | 63.1 |
| CANet | ResNet-50 | 51.2 | 57.3 | 51.5 | 52.1 | 53.0 | 55.3 | 59.7 | 55.5 | 53.7 | 56.0 | 57.7 | 62.4 | 58.5 | 58.8 | 59.3 |
| PANet | VGG-16 | 56.4 | 66.1 | 58.4 | 67.2 | 62.0 | 61.8 | 73.0 | 64.6 | 68.9 | 67.1 | 64.7 | 74.1 | 65.8 | 71.0 | 68.9 |
| PANet+ | HRNet-30 | 65.2 | 82.2 | 64.2 | 73.9 | 71.4 | 68.9 | 85.6 | 68.9 | 81.1 | 76.1 | 71.4 | 85.7 | 70.4 | 84.6 | 78.0 |
| DMML-Net | HRNet-30 | 68.6 | 82.2 | 66.8 | 78.4 | **74.0** | 74.8 | 87.6 | 74.3 | 87.5 | **81.1** | 79.7 | 90.3 | 78.4 | 90.1 | **84.6** |

6.5% higher in the five-shot setting, and 9.4% higher in the ten-shot setting. From the perspective of FB-F score evaluation, DMML-Net achieves the score 2.6% higher in the one-shot setting, 5.0% higher in the five-shot setting, and 6.6% higher in the ten-shot setting.

The superiority of the proposed DMML-Net can be analyzed from two aspects. First, the traditional metric-based methods usually manipulate prototype learning and metric comparison from a single scale. Considering the large intraclass variances of geographic objects, the metamodel exposes poor generalization ability when dealing with novel instances, while DMML-Net attempts to learn the prototypes from multiscale feature representations and dynamically reweights them for effective fusion, enhancing the robustness of the metamodel for intraclass variations. Second, the previous metric-based algorithms merely seek the shared feature representations and resolve all tasks utilizing the single static metric space. When dealing with learning tasks with large interclass distribution discrepancies, this training strategy exhibits poor learning efficiency. Contrarily, DMML-Net utilizes the metametric learning paradigm to conduct metric learning from both the task level and the metalevel. Benefiting from this pattern, both the task-specific and task-agnostic representations are effectively mined and jointly leveraged to boost the metric learning efficiency.

### *E. Ablation Study*

In this section, we conduct ablation experiments to explore the proposed DMML-Net and verify the effectiveness of our method.

*1) Effect of the Embedding Network Architecture in the Metric Learner:* Table VII shows the experimental results under the selection of different backbone networks. The comparison experiments are conducted on the mini-iSAID dataset with split-A chosen as novel classes. For the implementation of the VGG-16 [68], we inherit the architecture design in PANet [16]. In particular, only the first five blocks in the original VGG-16 are kept, and the dilated convolutions are utilized to manipulate the output stride into 8. When we use the modified VGG-16 as the backbone, our model still achieves relatively good results compared with the results of PANet in Table III: 3.7% higher on the mean-F score in the one-shot setting, and 10.4% higher on the mean-F score in the five-shot setting. For the implementation of the ResNet-50 [69], we still adopt the dilated convolutions techniques to set the

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TABLE VII
EFFECT OF THE ARCHITECTURE DESIGN FOR THE METRIC LEARNER

| Backbone | mean-F score (%) | | FB-F score (%) | | Params |
|---|---|---|---|---|---|
| | 1-shot | 5-shot | 1-shot | 5-shot | |
| VGG-16 | 56.8 | 68.4 | 79.7 | 84.3 | 14.71M |
| HRNet-18 | 57.8 | 70.7 | 78.2 | 84.6 | **9.56M** |
| ResNet-50 | 58.6 | 70.7 | 79.4 | 84.7 | 23.51M |
| HRNet-30 | **62.1** | **72.8** | **79.8** | **85.8** | 25.81M |

output stride of the network as 8 for fair comparison following the architecture design in DeepLab-v3 [70].

As shown in Table VII, with the increase in the depth and complexity of the embedding network, the performance of the model gains a steady improvement, indicating that the exploitation of deep features with high-level semantic representations brings a positive impact. Another phenomenon that can be observed is that the multiscale embedding architecture attains superior performance than the single-scale feature extractor with parameters at a similar magnitude. Specifically, compared with the results of VGG-16, HRNet-18 attains 1.0% higher on the mean-F score in the one-shot setting and 2.3% higher on the mean-F score in the five-shot setting. The mean-F score of HRNet-30 is 3.5% and 2.1% higher than the results of the ResNet-50 in one- and five-shot conditions. This fact can be analyzed as follows. The ensemble of feature descriptors from multiple scales helps enrich the detailed information of the item for intraclass variations. On the other hand, the prototypes obtained from the single scale lack flexible adaptability across the different geographic object categories. For example, vehicles usually only occupy a small area in the remote sensing imagery, while buildings generally possess enormous appearances. The optimal prototype representation for the corresponding tasks contains different biases on the scale of feature representations.

*2) Analysis of the Hyperparameter $\lambda$ for the Affinity-Based Fusion Mechanism:* We conduct the experiments to analyze the impact of the hyperparameter $\lambda$, which serves as the weight to control the credibility of the generated query pseudomasks in the affinity-based fusion mechanism. Specifically, the parameter analysis is conducted on the split-A of the mini-iSAID dataset in the five-shot settings. We experiment by selecting different values for $\lambda$ in the interval $[0, 2]$. Fig. 8 shows the concrete experimental results for the affinity-based fusion mechanism under the different values of the hyperparameter $\lambda$. When $\lambda$ is set as 0, the supervision on the generated pseudoquery mask is removed. The subsequent calculation for the affinity coefficients of each individual prototype may be disturbed by the irrelevant features introduced by the inaccurate query mask. With the increase in the values of $\lambda$, the corresponding segmentation accuracy attains the continuous increment benefiting from the additional constraints for the supervision of the pseudoquery mask. The model achieves the best accuracy when $\lambda$ is approximately 0.4. However, when the values of $\lambda$ become too large, the original objective for the supervision of the rectified prototypes can be diluted and lead to performance degradation.

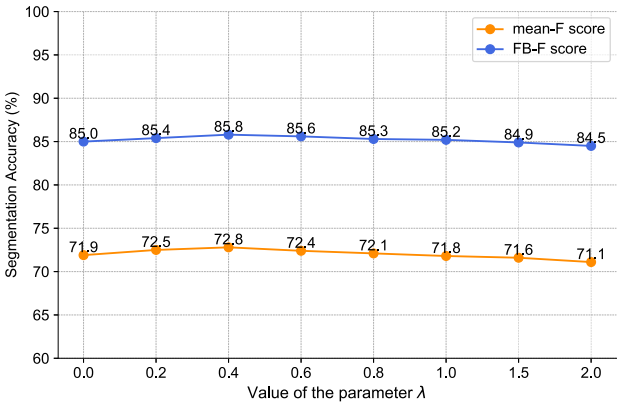



Fig. 8. Impact of the parameter $\lambda$ for the few-shot segmentation accuracy.

*3) Effect of the Affinity-Based Fusion Mechanism:* Fig. 9 shows the comparison results of different fusion methods on split-A of the mini-iSAID dataset under the conditions of five support samples available. The concrete fusion methods to be compared are listed as follows.

1) *Logical* OR*:* The $K$-shot segmentation task is decoupled into $K$ one-shot segmentation tasks to independently generate the prediction masks for the same query image. For the generation of the final segmentation result, the target pixel in the query image is classified into the foreground category if any one-shot segmenter predicts it as foreground.
2) *Logical* AND*:* This fusion method is similar to the logical OR fusion, and the final segmentation results are merged via applying the logic AND operation at each location of the query image.
3) *Mask Average:* This fusion technique still treats the $K$-shot segmentation task as $K$ one-shot segmentation tasks and generates the final result by performing averaging operations over the obtained segmentation probability maps.
4) *Prototype Average:* It is one of the most widely adopted fusion strategies, which conducts the average operation in the feature level and utilizes the averaged prototypes to approximate the ideal optimal prototypes.

As shown in Fig. 9, the fusion methods at the feature level generally yield better performance than image-level fusion methods. Besides, the feature-based fusion methods appear more efficient as the computation for metric classification only takes once or twice. The proposed affinity-based fusion method attains the most obvious increment without introducing any additional parameters to be optimized. This indicates the effectiveness of the proposed affinity-based fusion method.

*4) Analysis on the Selection of Distance Metrics in the Metric Space:* Fig. 10 shows the performance of the model under different criteria of distance metrics. Note that all the distance metrics adopt the same post-processing strategy, including dimensionality normalization [71] and temperature-based metric scaling [72], [73]. The experiments are conducted on the split-A of the mini-iSAID dataset under the conditions of one or five labeled samples available. The experimental results

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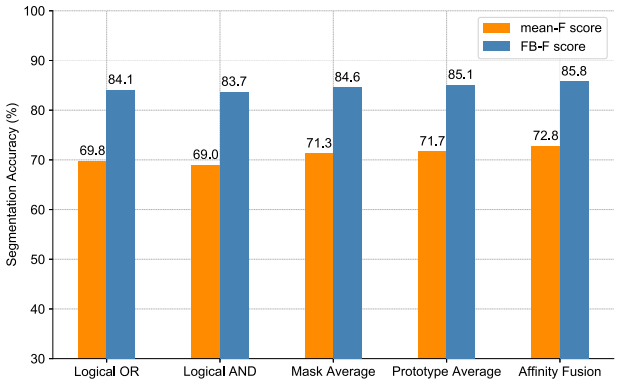



Fig. 9. Comparison of different fusion methods for five-shot segmentation on the mini-iSAID dataset.

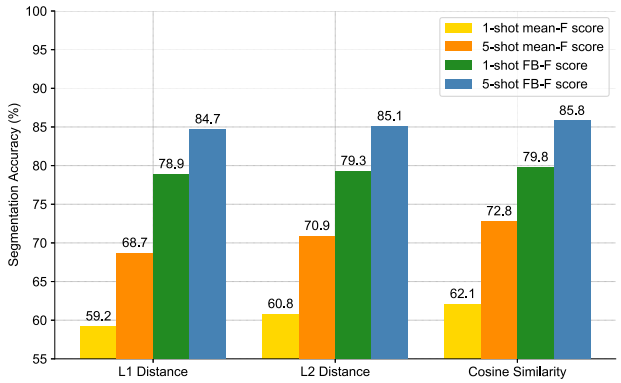



Fig. 10. Few-shot segmentation performance on the mini-iSAID dataset under different criteria of distance metrics.

demonstrate that the cosine similarity is a superior choice for the metric classification of the embeddings in the metric space compared with the L1 and L2 distances. We conjecture this is primarily because L1 and L2 distances mainly focus on the absolute distance across vectors. In comparison, cosine similarity measures the similarity of the vectors from the perspectives of intersection angle, which is especially suitable for the metric classification of 1-D embeddings with high dimensions.

*5) Effect of the Metametric Training Algorithm:* Table VIII shows the performance of the model under different training strategies. The comparison experiments are executed on the split-A of the mini-iSAID dataset under the conditions of one or five labeled samples available. For the implementation of the metric training algorithm, the entire training process still adopts the episode-based schema yet only involves a single metric learner. Each gradient update of the metric learner is directly calculated from the loss on the sampled minibatch of tasks.

As shown in Table VIII, the metric training strategy attains slightly better performance than the metametric training strategy in the conditions of non-fine-tuning. In combination with the fine-tuning techniques, the metametric training strategy achieves superior performance than the metric-based learning strategy. This experimental phenomenon can be analyzed

TABLE VIII
EFFECT OF THE METAMETRIC TRAINING ALGORITHM

| Methods | Finetuning | Mean-F score (%) | | FB-F score (%) | |
|---|---|---|---|---|---|
| | | 1-shot | 5-shot | 1-shot | 5-shot |
| Metric Training | × | 59.1 | 63.9 | 78.9 | 82.2 |
| | ✓ | 61.2 | 70.9 | 78.3 | 84.9 |
| Meta Metric Training | × | 58.3 | 62.8 | 78.2 | 81.0 |
| | ✓ | **62.1** | **72.8** | **79.8** | **85.8** |

from the optimization strategy within an episode. In metric learning, the episode is designed to mimic the scenarios where the model is directly applied with fine-tuning. Specifically, the model is directly optimized to learn the feature representations that are broadly beneficial across tasks. In comparison, the episode of metametric learning is designed to mimic the scenarios where the model is applied after fine-tuning. Concretely, the model is requested to find the parameters that can generate significant gains after the slight change.

Specific to geographic segmentation on remote sensing imagery where expose large discrepancy in the learning biases across classes, a metametric learning strategy is a superior option compared with the metric training strategy. Hence, the intraclass knowledge can be effectively exploited to assist the few-shot segmentation by task-specific adaptation. Furthermore, the metric space can be flexibly adaptable in accordance with the individual learning requirements of the tasks.

## V. CONCLUSION

This work introduces a novel few-shot learning framework, named DMML-Net, for the task of geographic object segmentation in remote sensing imagery. DMML-Net tackles the geographic object segmentation from a metric learning perspective and is capable of learning the metric space from very limited labeled samples for the pixelwise metric classification. In light of the characteristic of the large intraclass variances, DMML-Net makes improvements from two aspects. On the one hand, a deep feature pyramid comparison network is developed to conduct multiscale metric learning, increasing the robustness of the model to deal with the scale variations of geographic objects. On the other hand, an affinity-based fusion mechanism is proposed to adaptatively calibrate the deviation of the prototypes. Considering the impact of the large interclass distribution discrepancy, DMML-Net presents a novel metametric training algorithm to increase metric learning efficiency and enhance the scalability of the model for fast adaption to various tasks. The experimental results on two benchmark datasets demonstrate the validity and the superiority of our proposed DMML-Net in low-shot regimes.

As part of future work, since this work is the initial exploration for the task of few-shot geographic object segmentation in the remote sensing field, we are interested in investigating how to further enhance the intraclass generalization and interclass universality of the metamodel. In addition, we will further study the theoretical guarantees for the corresponding few-shot learning research.

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